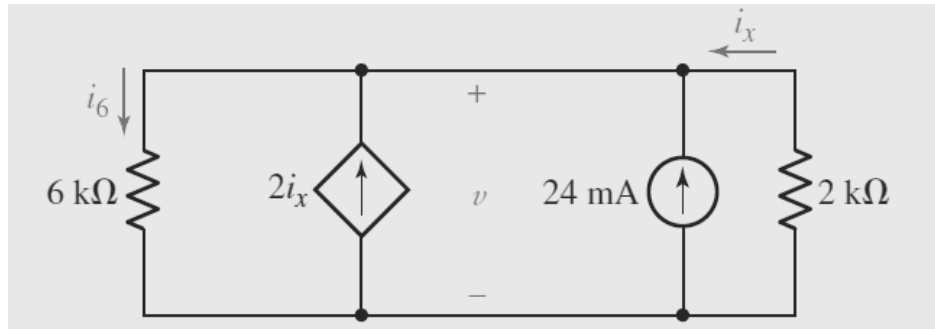


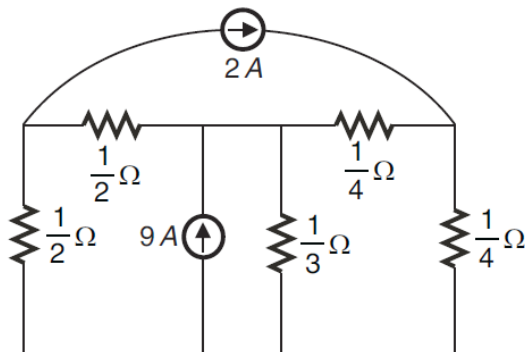
EC1101 – Tutorial Sheet 2

- 1 Determine the value of v and the power supplied by the independent current source.



$$v = 14.4 \text{ V}; p_{24} = 0.3456 \text{ W}$$

- 2 Find all the node voltages

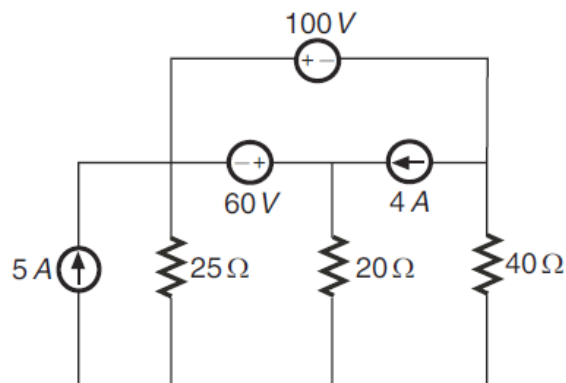


$$V_1 = \frac{\Delta'_1}{\Delta'} = \frac{48}{192} = 0.25 \text{ V}$$

$$V_2 = \frac{\Delta'_2}{\Delta'} = \frac{288}{192} = 1.5 \text{ V}$$

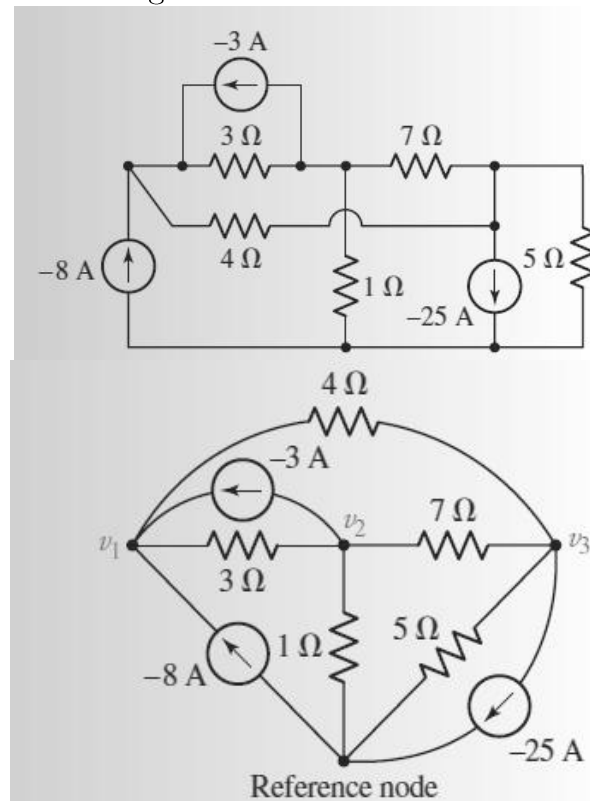
$$V_3 = \frac{\Delta'_3}{\Delta'} = \frac{192}{192} = 1 \text{ V}$$

- 3 Find all the node voltages.



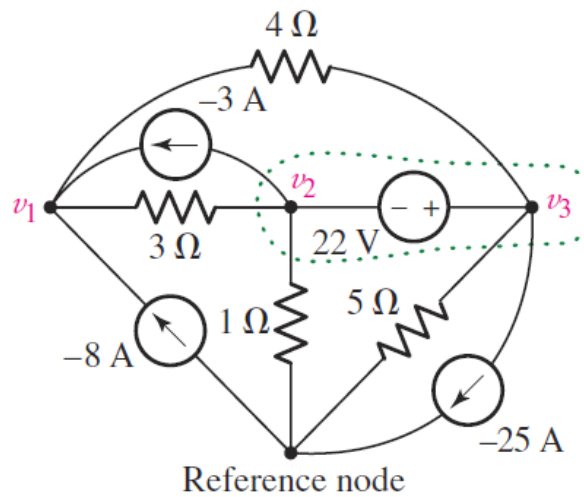
$$V_1 = 39.13 \text{ V} \quad V_3 = -60.87 \text{ V}$$

-
- 4 Determine the nodal voltages as referenced to the bottom node



V1: 5.412 V, V2: 7.736 V, V3: 46.32 V

-
- 5 Determine the value of node voltage v_1 in the circuit below.



V1: 1.071 V
